UBT407: TRANSDUCERS AND BIOSENSORS

L	T	P	Cr
3	0	2	4.0

Course Objective: The course aims to impart knowledge on basic concepts of transducers and acquaint the students with different types of electrodes used in bio-potential recording. The course will also provide understanding of biosensors, optical and ultrasonic sensors.

Syllabus

Physiological Transducers: Transducers in general, active and passive transducers, pressure transducers, catheter tip pressure transducers, temperature transducers, pulse sensors, respiration sensors, digital transducers, selection criteria for transducers.

Bioelectric potentials/Physiological signals: Action potentials and impulse propagation, origin of bioelectric signals, electrode theory, types of electrodes, selection criteria for electrodes recording electrodes and skin-contact impedances, electrical conductivity and microelectrodes, pulse, temperature, pressure and repression sensors.

Biosensors: Benefits of biosensors, Types of biosensors, potentiometry, Bio-chemical sensors, chemical potential and equilibrium - some famous examples - electrochemical cell at equilibrium - Nernst equation - pH electrode - Ion-sensitive electrodes, voltammetry, amperometry, conductimetry.

Ultrasonic, Optical & Laser biosensors: Basics of ultrasound, theory, characteristics, design, applications in medical science for diagnostic and therapeutic, Optical fiber sensor, Polarization, Refractive index, Light scattering, micro-opto- electromechanical system [MOEMS], Laser in industry.

Signal processing: Introduction to biomedical signal processing and analysis; Wheatstone bridge, Bioelectric amplifiers, instrumentation amplifier, Introduction to active filters, First order, second order and higher order filters, Modulation and demodulation.

Laboratory Work (if applicable):

Experiments based on strain gauge, LVDT, capacitance, photoelectric, piezoelectric and temperature. Also, experiments for digital sensor, LDR, resistivity measurement.

Course Learning Objectives (CLO)
The students will be able to:

1. Explain basic concepts of transducers
2. Elucidate different types of electrodes used in bio-potential recording

3. Differentiate biosensors, optical and ultrasonic sensors
4. Analyze, formulate and select suitable sensor/biosensor.

Text Books

1. Cromwell, L. and Weibell, F.J. and Pfeiffer, E.A., Biomedical Instrumentation and Measurement, Dorling Kingsley (2006) 2nd ed.
2. Carr, J.J. and Brown, J.M., Introduction to Biomedical Equipment Technology, Prentice Hall (2000) 4th ed.

Reference Books

1. A.K. Sawhney and Puneet Sawhney, A Course in Electrical and Electronic Measurements and Instrumentation, Dhanpat Rai, 2014
2. Florinel-Gabriel Banica, Chemical Sensors and Biosensors: Fundamentals and Applications, Wiley, 2012

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	35
3.	Sessionals (May include assignments/quizzes)	40